

Functional Description

Gas engine-generator sets

MTU 8V4000 GS – GG08V4000A1/A2 with engines 8V4000L32/L32FB/L33 and 8V4000L64

MTU 12V4000 GS – GG12V4000A1/A2 with engines 12V4000L32/L32FB/L33 and 12V4000L64

MTU 16V4000 GS – GG16V4000A1/A2 with engines 16V4000L32/L32FB/L33 and 16V4000L64

MTU 20V4000 GS – GG20V4000A1/A2 with engines 20V4000L32/L32FB/L33 and 20V4000L64

MS13085/00E

Name	Model number	Engine model	Application group
MTU 8V4000 GS	GG08V4000A1	8V4000L32 8V4000L32FB 8V4000L33 8V4000L64	3A, continuous operation, unrestricted
	GG08V4000A2	8V4000L32 8V4000L32FB 8V4000L33 8V4000L64	3A, continuous operation, unrestricted
MTU 12V4000 GS	GG12V4000A1	12V4000L32 12V4000L32FB 12V4000L33 12V4000L64	3A, continuous operation, unrestricted
	GG12V4000A2	12V4000L32 12V4000L32FB 12V4000L33 12V4000L64	3A, continuous operation, unrestricted
MTU 16V4000 GS	GG16V4000A1	16V4000L32 16V4000L32FB 16V4000L33 16V4000L64	3A, continuous operation, unrestricted
	GG16V4000A2	16V4000L32 16V4000L32FB 16V4000L33 16V4000L64	3A, continuous operation, unrestricted
MTU 20V4000 GS	GG20V4000A1	20V4000L32 20V4000L32FB 20V4000L33 20V4000L64	3A, continuous operation, unrestricted
	GG20V4000A2	20V4000L32 20V4000L32FB 20V4000L33 20V4000L64	3A, continuous operation, unrestricted

Table 1: Applicability

© 2017 MTU Onsite Energy GmbH, Augsburg

The original document was created in the German language.

This publication including all of its parts is protected by copyright. Each instance of use requires the prior written consent of MTU Onsite Energy GmbH. This applies in particular to duplication, dissemination, editing, translation, microfilming, and storage and/or processing in electronic systems, including databases and online services.

All of the information presented in this publication was current and up-to-date at the time of publishing. MTU Onsite Energy GmbH reserves the right to make changes to, delete, or supplement the provided information and data as needed.

Table of Contents

1 Engine-Generator Set		7.2 Catalytic converter	53
1.1 General description	4	7.3 Exhaust silencer	54
1.2 Engine-generator set – Application group	5	8 Lube Oil System	
1.3 Engine-generator set – Nameplate and designation	6	8.1 Lube oil system	55
1.4 Engine-generator set – Standard scope of delivery	9	8.2 Oil priming pump	57
1.5 Engine-generator set – Optional scope of delivery	18	9 Cooling System and Heating Water System	
2 Engine		9.1 Cooling system	58
2.1 Engine – Overview	19	9.2 Heat module	61
2.2 Engine side and cylinder designations	26	10 Base Frame	
2.3 Sensors – Overview	27	10.1 Mounting	65
2.4 Mixture formation, turbocharging and exhaust system	31	11 Monitoring, Control and Regulation Equipment	
2.5 Lube oil system	34	11.1 Gas engine phase 4 system – Overview	67
2.6 Cooling system	36	11.2 Purpose of the units	69
3 Gearbox / Transmission		12 Electrical System	
3.1 Gearbox	41	12.1 Starter battery	71
4 Generator		12.2 Battery charger	72
4.1 Generator – General information	43	13 Gas Alarm and Fire Alarm	
4.2 Coupling	44	13.1 Gas alarm system	73
5 Monitoring and Control		13.2 Fire detection system	74
5.1 MIP and MMC control system	45	14 Technical Data	
6 Charge-air System and Gas System		14.1 Technical data	76
6.1 Compressor	46	15 Appendix A	
6.2 Gas train	47	15.1 Abbreviations	77
6.3 Air filter	49	16 Appendix B	
7 Exhaust System		16.1 Index	79
7.1 Exhaust gas heat exchanger	51		

1 Engine-Generator Set

1.1 General description

The engine-generator set comprises the following main assemblies:

- Engine
- Generator and coupling
- Gearbox and coupling (60-Hz gensets)
- Gas train
- Base frame
- Heat extraction (with versions GR and GC)
- MMC (MTU Module Control) for plant control, closed-loop control, diagnostics (optional)
- MIP (MTU Interface Panel) for engine control, closed-loop control, diagnostics

In the combustion engine, the energy contained in the fuel is converted into mechanical work and heat.

The generator, which converts the mechanical work into electrical energy, is flanged-mounted on the engine via a resilient coupling. The generator is installed on two slide rails, which allow the generator to be shifted when the coupling element is being replaced. The engine and generator are connected to the base frame via resilient, vibration-damped elements.

All closed-loop controls, control systems and monitoring systems, including communication options to and from the system, are implemented in the hardware and software in the control cabinet.

1.2 Engine-generator set – Application group

The engine-generator set consists of a gas engine and a generator mounted on a common baseframe. Upon demand, the engine will start and drives the generator to produce electrical power.

Application group 3A – Continuous operation, unrestricted (Continuous Power)

Engine-generator sets for unlimited continuous power supply are used for:

- Base load generation
- Part of a power supply or the public grid

Continuous operation	Application group 3A
Operating mode	Continuous operation, unrestricted
Rating definition	10% overload capacity (ICXN)
Load factor	< 100 %
Operating hours	Unrestricted

Benefits

- Wide range of standardized engine-generator sets to meet customer requirements with regard to power, emission and other characteristics.
- Possibility to choose from different component designs (e.g. generator) and to select options (e.g. heat module)
- Most recent gas engine technology.
- Main components with state-of-the-art technology for high electrical and thermal efficiency and long service life.

1.3 Engine-generator set – Nameplate and designation

Nameplates on engine-generator set

The genset has the following nameplates:

- Nameplate of engine-generator set
- Engine nameplate
- Nameplate of generator

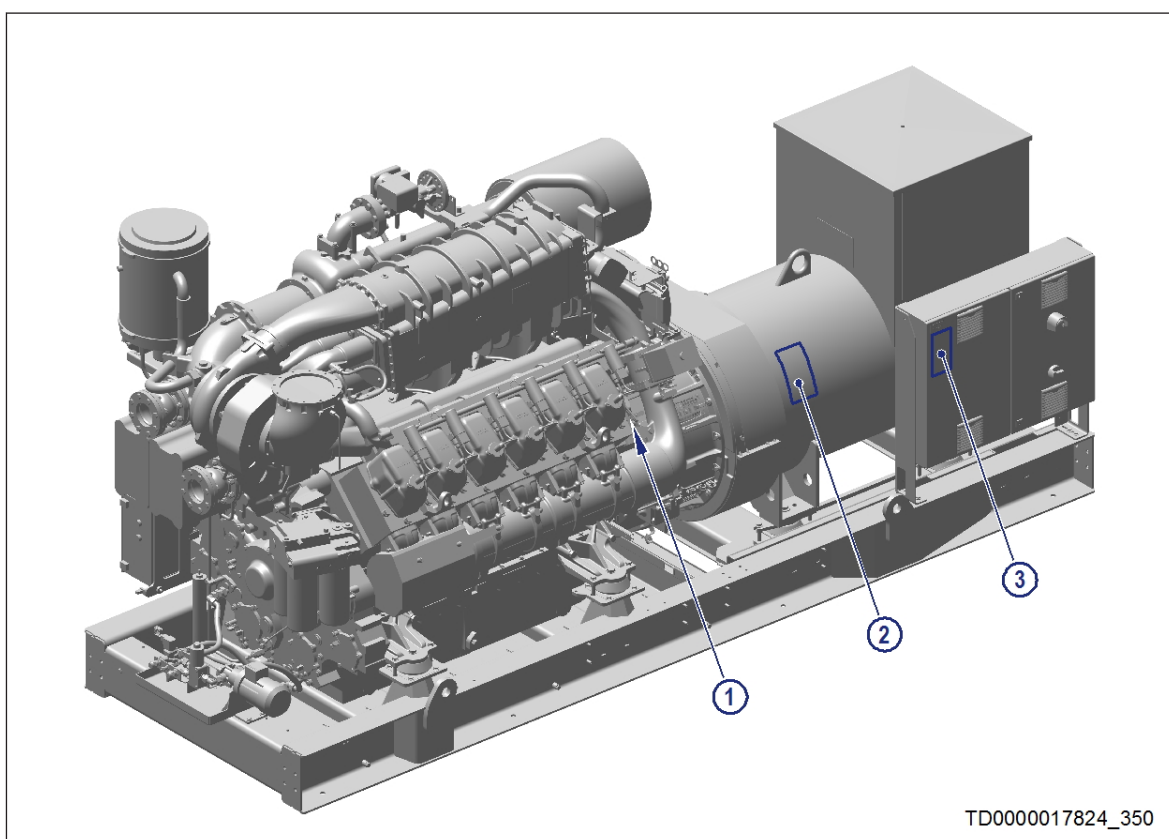


Figure 1: Fitting position of nameplates (generator and engine-generator set)

1 Engine nameplate

2 Nameplate of generator

3 Nameplate of engine-generator set

Explanation of product name

Example: MTU 12V4000 GS

MTU	Manufacturer: MTU Onsite Energy
12	Number of cylinders of the engine
V	Cylinder arrangement of the engine :Vee arrangement
4000	MTU Series 4000
GS	Gas System

Explanation of model number

Example: GG12V4000A1

GG	Product type: Gaset
12	Number of cylinders of the engine
V	Cylinder arrangement

Example:GG12V4000A1	
4000	4000 series
A1	Design/version

Overview of application groups on nameplate

Specification on the nameplate	Application group
CONTINUOUS POWER	3A – Continuous operation, unrestricted
PRIME POWER	3B – Continuous operation, variable load, ICXN
LTP GRID STABILITY	3G – Continuous operation, time limited, ICXN
DATA CENTER CONTINUOUS	3F – Standby operation mode, unlimited, ICXN
MISSION CRITICAL STANDBY	3H – Standby operation mode, special qualifications

Engine-generator set – nameplate and designation

The nameplate is a vertical rectangular plate with the MTU Onsite Energy logo at the top right. It contains the following fields and callouts:

- 1** points to the top section: **Stromerzeugungsaggregat** / **Genset**, followed by ISO standard power ratings and environmental conditions (air pressure, temperature, humidity).
- 2** points to the middle section: **Wärmeauskopplung nach DIN 6280** / **Heat extraction to DIN 6280**, followed by thermal output and heating water temperature/pressure.
- 3** points to the bottom left: **MTU Onsite Energy GmbH**, **Dasinger Str. 11**, **D-86165 Augsburg**, and the serial number **XZG3900400001**.
- 4** points to the **Typ** (Type) field.
- 5** points to the **Modul- / Aggregat- Nr.** (Module / Genset No.) field.
- 6** points to the **Projekt Nr.** (Project No.) field.
- 7** points to the **Baujahr** (Year of Manufacture) field.
- 8** points to the **Gewicht mit / ohne Betriebsstoffe [kg]** (Weight with / without fluids and lubricants [kg]) field.
- 9** points to the **Elektrische Leistung COP [kW]** (Electrical Power COP [kW]) field.
- 10** points to the **Performance - Class** field.
- 11** points to the **Nennspannung [V]** (Nominal Voltage [V]) field.
- 12** points to the **Nennfrequenz [Hz]** (Nominal Frequency [Hz]) field.
- 13** points to the **Max. Außentemperatur [°C]** (Max. outdoor temperature [°C]) field.
- 14** points to the **Nennleistungsfaktor cos φ induktiv / kapazitiv** (Nominal Rated Power Factor cos φ inductive / capacitive) field.
- 15** points to the **Max. Aufstellhöhe [m ü.NN.]** (Max. installation height [m a.s.l.]) field.
- 16** points to the **Zul. Heizwassertemperatur [°C]** (Admissible heating water temperature [°C] min./max.) field.
- 17** points to the **Zul. Betriebsdruck im Heizwasserkreis** (Admissible operating pressure in heating water circuit) field.
- 18** points to the **CE** mark and **Made in Germany** text.

The number **00002761** is printed at the bottom right of the nameplate.

Figure 2: Nameplate of engine-generator set

Item	Designation
1	Power ratings as per ISO standard power ISO 3046-1 no overload capability Air pressure (total pressure) $p_r = 100 \text{ kPa}$ (100 above sea level) Air temperature (at air filter inlet) $T_r = 298 \text{ K}$ ($t_r = 25 \text{ °C}$) Relative air humidity $\phi = 30\%$
2	Heat supply from cogeneration as per DIN 6280
3	MTU Onsite Energy Dasinger Str. 11 D-86165 Augsburg
4	Type
5	Module / genset no.

Item	Designation
6	Project no.
7	Year of construction
8	Weight with / without fluids and lubricants [kg]
9	Electrical output COP [KW]
10	Performance class
11	Rated voltage [V]
12	Rated frequency [Hz]
13	Max. outside temperature [°C]
14	Rated power factor $\cos \phi$ inductive / capacitive
15	Max. site altitude [m above sea level]
16	Thermal rated power [kW]
17	Perm. heating water temperature [°C]
18	Perm. operating pressure in heating water circuit

1.4 Engine-generator set – Standard scope of delivery

Overview of engine-generator sets 50 Hz

Genset 20V4000Lxy

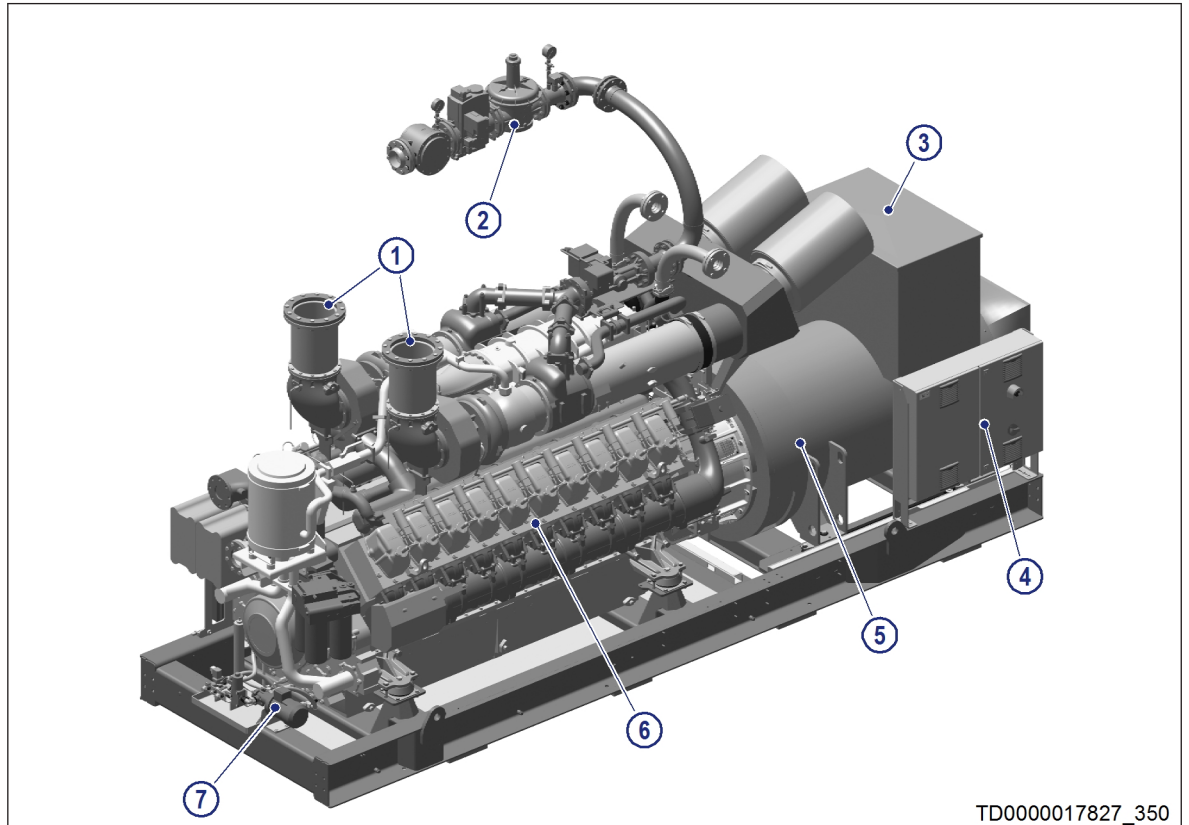


Figure 3: Engine-generator set 20V – engine side

- | | | |
|--------------------------|-------------------|-----------------|
| 1 Exhaust outlet | 4 Control cabinet | 7 Lube oil pump |
| 2 Gas train | 5 Generator | |
| 3 Generator terminal box | 6 Engine | |

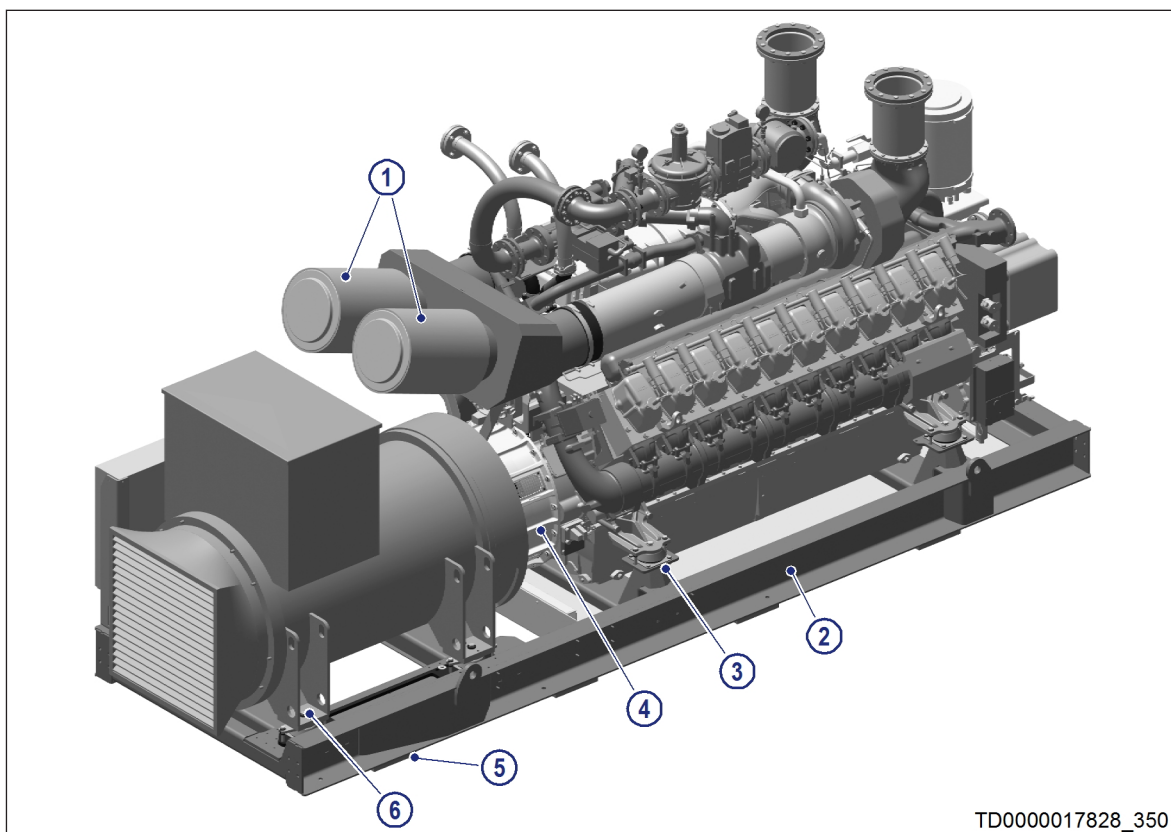


Figure 4: Engine-generator set 20V – generator side

- | | | |
|--------------|-------------------|---------------------|
| 1 Air filter | 3 Engine mounting | 5 Damping plate |
| 2 Base frame | 4 Coupling | 6 Generator bearing |